Ratanakmuny NOUV

Master's Student in Data Science | Applied Mathematics Engineer

Seeking a 4–6 month end-of-studies internship in Data Science / AI

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Education

Université Paris-Saclay, Évry *Sep 2025 – Present*

Master's degree in Data Science — Health, Insurance, Finance track.

ENSIIE, Évry *Sep 2022 – Oct 2025*

École Nationale Supérieure d'Informatique pour l'Industrie et l'Entreprise.

Engineering degree (Master's equivalent) in Applied Mathematics.

Institut de Technologie du Cambodge (ITC), Phnom Penh *Nov 2019 – Sep 2025*

Engineering degree in Computer Engineering — ITC/ENSIIE double degree.

Professional Experience

Paris Partners Softwares (PPS) — Innovation Department *Mar 2025 – Sep 2025*

Research Intern — AI/ML Engineer: AI-assisted penetration testing

- Researched and developed AI-assisted pentesting techniques for web applications.
- Surveyed and benchmarked vulnerability-detection and automated-scanning tools.
- Designed internal penetration-testing workflows.
- Hands-on experience with security testing methodologies and reporting.

AI Farm Co., Ltd. — Development Department *Jun 2024 – Aug 2024*

Project Manager, Business Analyst & Backend Developer

- Led cross-functional teams on a complex, high-impact project.
- Captured business requirements and developed backend solutions.
- Collaborated with the data-engineering team within the data warehouse environment.
- Designed and optimized the transaction flow for seamless data integration and accessibility.

Paris Partners Softwares (PPS) — Development Department *May 2023 – Jul 2023*

Backend Developer

- Developed REST APIs and logistics features consumed by the frontend.
- Backend stack: Laravel (PHP).

Academic Projects

Research Paper Assistant: RAG-Based Question Answering *Python, LangChain, FastAPI*

- Building a retrieval-augmented generation system for grounded question answering over a corpus of research papers.
- Dense retrieval via sentence-transformer embeddings stored in ChromaDB, served through a FastAPI layer with Docker.
- Prompt-engineering layer integrating LLM API calls for citation-aware answers grounded in retrieved passages.

Dimensionality Reduction: ICA and EM-ICA extension *Python*

- Implemented 4 ICA algorithms (Infomax, FastICA, SGD-ICA, EM-ICA) in a unified gradient-descent framework on whitened data.
- **Original contribution — EM-ICA**: learning the score function via a Gaussian Mixture Model (GMM), replacing the fixed kurtosis switch.
- Evaluation on 5 datasets (synthetic, audio, images, real photographs, real EEG) using the Amari index, error bars over 10 runs, noise-robustness analysis.
- **Key result**: EM-ICA achieves 2.5× better separation on real photographs (Amari = 0.034 vs 0.086) thanks to learned source densities.

Machine Learning in Quantitative Finance: multi-fidelity Gaussian-process regression for market risk *Python*

- Reproduced and extended Lehdili, Oswald & Nguyen (2025): pricing-engine surrogate via Gaussian processes for VaR/ES computation under FRTB.
- From-scratch implementation of the Kennedy & O'Hagan AR(1) co-kriging model with joint marginal-likelihood optimization (Cholesky, L-BFGS-B).

- Multi-fidelity data generation (Monte Carlo, 100k vs 100 paths) with Common Random Numbers (CRN) on a portfolio of geometric options on 5 correlated assets.
- Comparison of standard GPR vs mGPR at equal HF budget: accuracy gain (MAPE, VaR, ES) when expensive training points are scarce.

Algorithmics: Bayesian-network structure learning *R / Rcpp*

- Learned the structure of a Bayesian network from data (DAG construction).
- Implemented and compared search methods (Hill-Climbing, Tabu Search, Branch-and-Bound) with the BIC score.
- Handled both continuous and discrete cases (Gaussian and multinomial nodes) with associated parameter estimation.
- Optimized runtime via Rcpp and analyzed performance (score, complexity, convergence).

Financial Econometrics: volatility modeling and risk measurement *R*

- Time-series analysis of financial returns and study of stylized facts (volatility clustering, asymmetry, fat tails).
- Conditional-volatility modeling via ARCH/GARCH and parameter interpretation (persistence, shocks).
- Estimation, diagnostics and model comparison (residual analysis, backtesting).
- Estimation vs forecasting perspective on volatility measures.

Survival Analysis: oncology prognosis on the METABRIC dataset *R*

- Right-censored survival data analysis (time-to-event, status indicator).
- Nonparametric estimation of survival curves (Kaplan-Meier) and group comparison.
- Hazard modeling via the Cox proportional-hazards model, hazard-ratio interpretation and PH-assumption checks.
- Predictive survival models (Random Survival Forest), evaluation via C-index and Brier score / IBS.

DataCamp / RAMP Challenge: cell-type classification on single-cell RNA-seq *Python*

- Supervised classification of 4 cell types from sparse gene-expression matrices.
- Preprocessing and feature-engineering pipelines (normalization, standardization, PCA, count-adapted TF-IDF).
- Comparison of models (Logistic Regression, Random Forest, Gradient Boosting, LightGBM, stacking) on balanced accuracy.
- Iterative performance optimization (CV, error analysis, hyperparameter tuning) tracked on the RAMP server.

Deep Learning: MNIST classification and semi-supervised GAN *Python*

- Digit classification (0–9) on MNIST with a CNN baseline.
- Implemented techniques from *Improved Techniques for Training GANs* (Salimans et al., 2016): feature matching, minibatch discrimination, one-sided label smoothing.
- Semi-supervised setup: added a $(K+1)$-th "fake" class, training on labeled and unlabeled data.
- Compared CNN baseline vs improved approach (balanced accuracy, confusion matrix).

Deep Learning: Twitter sentiment analysis *Python*

- Sentiment classification (positive, negative, neutral) on noisy tweets.
- Preprocessing: tokenization, stop-word removal, Word2Vec embeddings.
- Neural models: LSTM, BiLSTM with attention, CNN.
- Evaluation: precision, recall, F1.

Statistical Modeling: opinion-trend analysis *R*

- Time-series analysis of opinion data.
- Linear and polynomial regressions, time-series modeling.
- Hypothesis testing and confidence-interval estimation.
- Visualizations and statistical summaries.

Probabilistic Graphical Models: Restricted Boltzmann Machines *Python*

- Implemented RBMs to model complex distributions.
- Unsupervised learning via contrastive divergence.
- Analysis of hidden representations.
- Evaluation for dimensionality reduction and reconstruction.

Machine Learning: Greenland ice quantities from infrasound signals *Python*

- Binary classification (low/high ice) from infrasound signals and environmental covariates.

- Log transforms, standardization and SMOTE to handle class imbalance.
- Random Forest, Logistic Regression, SVM, Gradient Boosting with polynomial feature engineering.

Machine Learning: OptDigits classification project *Python*

- K-means clustering of handwritten digits from numerical features.
- Preprocessing to improve cluster quality.
- Performance analysis and interpretation of obtained clusters.

Kaggle: Spaceship Titanic *Python*

- Predicting passenger transport based on personal data.
- Data preprocessing and feature engineering.
- Model comparison (Logistic Regression, SVM, Random Forest, CatBoost).

Molecule classification and impact on the circadian rhythm *R*

- Classification model to identify the influence of molecules on circadian-rhythm period.
- Statistical analysis of molecular features to extract relevant attributes.
- Evaluation and optimization of the predictive model.

IMT Starter Challenge — Entrepreneurship Projects 2024 *Innovation*

- Innovated on an ICT-based start-up project (small start-up track).

Volunteering

- Physics tutoring for first-year engineering students at ITC.

Skills

Programming: Python, R, SQL, C / C++.
Machine Learning: scikit-learn, PyTorch, TensorFlow, Keras, LangChain.
Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn.
Web & Databases: Flask, SQLAlchemy, PostgreSQL, SQLite, HTML, CSS, JavaScript.
Tools: Git, Docker, LaTeX, Kali Linux, Canva, Microsoft & Google Office.

Languages

- Khmer — native.
- English — C1 (Cambridge).
- French — B1 (TCF).

Personal Interests

Comfortable working in teams or autonomously; calm, dynamic, and easy-going.